Supplementary Material

Reassessing Photoplethysmography-Based Blood Glucose Estimation: Patient-Independent Validation, Optical Modeling, and Embedded Implementation

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Tables and figures referred to without the prefix S are those of the main article. References cited here are listed at the end of this document.

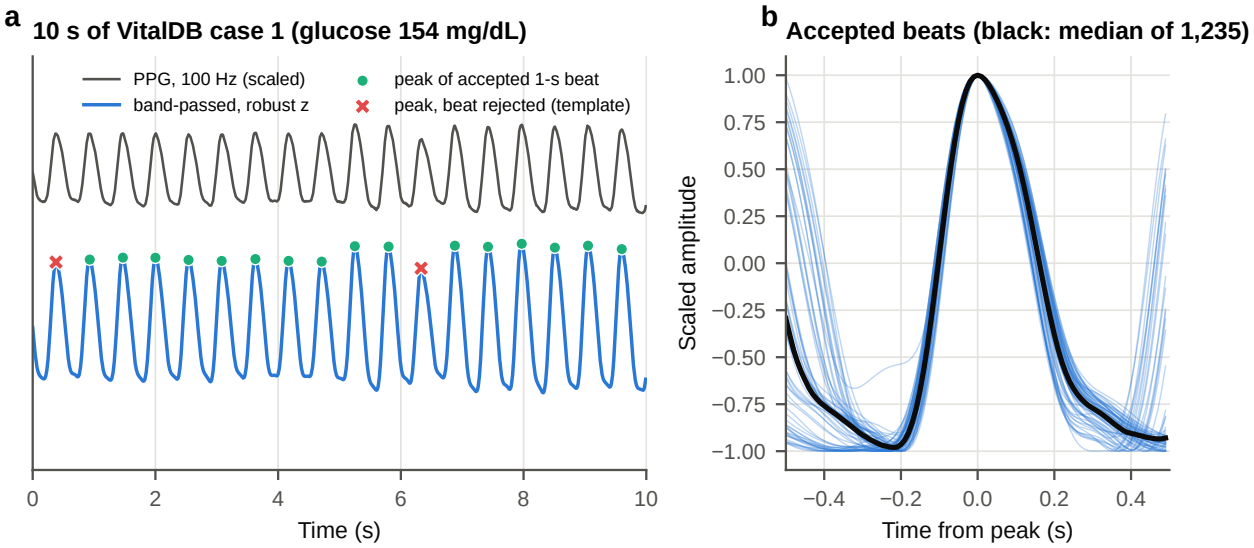


Figure S1. Preprocessing on real data: the 16-min finger PPG window around the first glucose draw of VitalDB case 1 (154 mg/dL), decimated to 100 Hz. (a) Ten seconds of this window (samples 9,500 to 10,500): the PPG and the band-passed, robust z-scored signal with the detected peaks; the two traces are scaled separately and offset vertically for display. Green circles mark the centers of accepted 1-s beats; the two red crosses mark peaks rejected by the template check (cosine similarity to the mean beat below 0.85). Over the whole window 1,352 peaks were detected and 1,235 beats accepted (91%). (b) Sixty accepted beats drawn at random from the window, each scaled to $[-1, 1]$, and the median of all 1,235.

Table S1. Origin of the optical parameters used in the simulation.

Parameter	Value and basis	Source
Glucose scattering coefficient	$(1/\mu_s)\,\partial\mu_s/\partial C_g = -0.29\%\,(\mathrm{mmol/L})^{-1}$ ($\partial\mu_s/\partial C_g \approx -0.041$ mm^{-1} per mmol/L), measured in 25 oral glucose tolerance tests at 1314 nm and extended over 1000 to 1700 nm with Mie theory	Ge et al. [1]
Independent estimate	-0.22% per mmol/L near 1300 nm (optical coherence tomography in animals), as cited by Ge et al.	Larin et al. [2]
Geometry	Three layers (0.1 / 1.0 / 10 mm), 0.2 mm glass window, detector rings at 1.7 to 2.9 mm; our model reproduces this geometry	Ge et al. [1]
Glucose absorption, 1000 to 1700 nm	$\partial\mu_a/\partial C_g$ spectrum used by Ge et al.	Ge et al. [1]; Amerov et al. [3]
Glucose bands below 1000 nm	CH band near 910 nm; OH bands at 960 and 985 nm change with temperature in the same way as with sugar concentration	Golic et al. [4]
Skin layers, 1000 to 1700 nm	Absorption and reduced scattering spectra of epidermis, dermis and subcutaneous fat, digitized	Iino et al. [5]
Skin layers, 660 to 1000 nm	μ_a from volume fractions of blood (with saturation), water, fat and melanosomes plus a layer baseline; μ_a of melanosomes $1.7 \times 10^{11}\lambda^{-3.48}$ mm^{-1} (λ in nm); $\mu_s' = a(\lambda/500\,\mathrm{nm})^{-b}$ with (a,b) = (6.87, 1.161), (4.53, 1.292) and (1.84, 0.672) mm^{-1} for epidermis, dermis and fat; compared with measured spectra [6, 7]	Jacques [8]; Bosschaart et al. [9]
Alternative model, 600 to 1700 nm	Composition-based μ_a of all layers; μ_s' measured ex vivo, power-law fit above 1330 nm; g held at 0.91 above 1000 nm; subcutaneous composition; glucose molar absorptivity at 660 and 940 nm	Salomatina et al. [10]; van Gemert et al. [11]; Ganesan et al. [12]; Haque et al. [13]
Conventions	Absorbance $A = -\ln R$ (natural logarithm); glucose sensitivities per mmol/L (1 mmol/L = 18 mg/dL)	–
Water absorption	Liquid water at 25 °C (reported run); Kou et al. in the earlier 1000–1700 nm run, with the same results	Segelstein [14]; Kou et al. [15]

Table S2. Robustness of the deep-network results to the training seed and to two training settings (per-reading scores on VitalDB, fixed folds). Seed rows give mean ± SD across runs.

Model	Split	Runs	RMSE (mg/dL)	R^2	Pearson r	SD of est. (mg/dL)	Clarke A (%)
ResNet34-1D, 1-s beats	Segment	2 seeds	31.53 ± 0.23	0.221 ± 0.011	0.563 ± 0.030	9.2 ± 1.6	57.2 ± 1.0
ResNet34-1D, 1-s beats	Person	3 seeds	34.90 ± 0.06	0.046 ± 0.003	0.229 ± 0.002	5.3 ± 0.7	54.4 ± 0.3
TinyCNN, 1-s beats	Person	5 seeds	34.67 ± 0.06	0.058 ± 0.003	0.245 ± 0.008	7.6 ± 0.2	55.4 ± 0.3
ResNet34-1D, lr 3×10^{-4}	Person	1	34.87	0.047	0.234	5.3	54.3
ResNet34-1D, 96 beats per reading	Person	1	34.98	0.042	0.221	4.9	53.7
Age, sex, BMI (ridge)	Person	–	35.0	0.041	–	–	54.4

Seed runs include the original run of Table 4 (seed 42) plus seeds 1–4 (TinyCNN), 1–2 (ResNet34-1D, person split) or 1 (ResNet34-1D, segment-level split). lr: learning rate. The variants use seed 42 and otherwise the settings of Table 3. SD of estimates: spread of the per-reading predictions; laboratory SD 35.7 mg/dL.

Table S3. PPG computed from the centered window or only from signal before the draw (person-level 5-fold). PPG gain: paired change in RMSE against the same model without PPG, $\Delta\text{RMSE} = \text{RMSE}_{\text{with PPG}} - \text{RMSE}_{\text{without PPG}}$; a negative value means that PPG improves accuracy. The centered-window rows are the models of Tables 6 and 8.

Setting	PPG window	RMSE (mg/dL)	R^2	PPG gain [95% CI]
No calibration	none (clinical + time)	32.93	0.150	–
(7,224 readings)	centered ±8 min	31.82	0.207	−1.11 [−1.50, −0.74]
	8 min before	32.12	0.192	−0.81 [−1.17, −0.45]
	2 min before	32.09	0.193	−0.83 [−1.23, −0.40]
	1 min before	32.03	0.196	−0.90 [−1.26, −0.50]
One calibration value	none (calibration + clinical + time)	32.97	0.228	–
(4,233 readings)	centered ±8 min	31.25	0.306	−1.72 [−2.74, −0.84]
	8 min before	31.43	0.298	−1.53 [−2.73, −0.56]
	2 min before	31.34	0.302	−1.63 [−3.14, −0.53]
	1 min before	31.40	0.299	−1.57 [−3.03, −0.57]

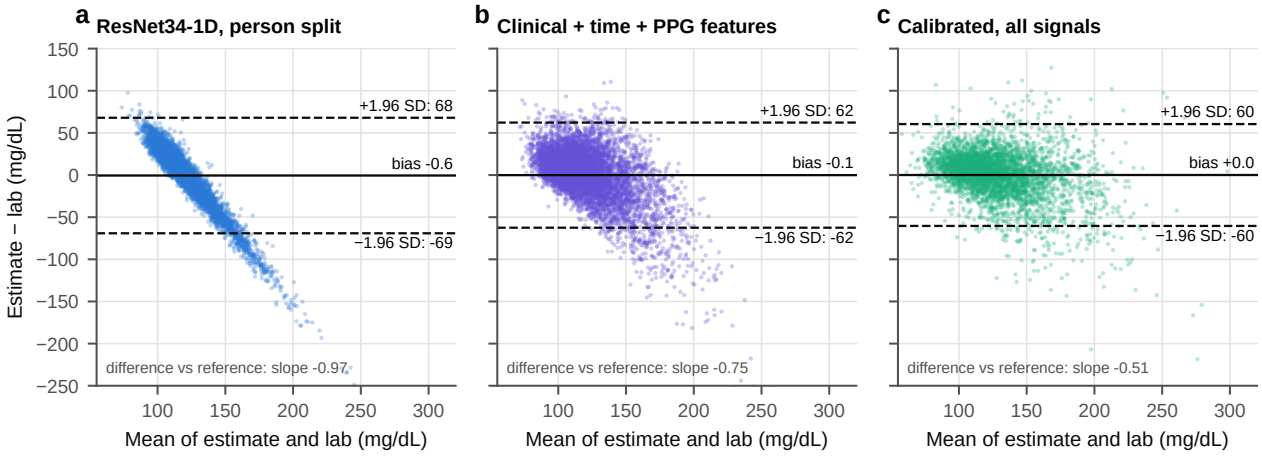


Figure S2. Bland–Altman plots, one point per reading: (a) ResNet34-1D with a person-level split, (b) clinical context, time since anesthesia and PPG features, (c) the calibrated model with all signals. Solid line: mean difference (bias); dashed lines: 95% limits of agreement. The annotated slopes were obtained by regressing the difference (estimate − laboratory) on the laboratory reference value, not on the mean shown on the horizontal axis; they measure regression toward the mean (−1 for a constant estimate, 0 for no proportional error).

Table S4. Accuracy by subgroup (person-level 5-fold): RMSE (mg/dL) and bias (estimate − laboratory, mg/dL).

		Training-set mean		ResNet34-1D		Clinical + time + PPG		Calibrated, all signals	
Subgroup	Readings[a]	RMSE	Bias	RMSE	Bias	RMSE	Bias	RMSE	Bias
All readings	7,224 / 4,233	35.7	0	35.0	−1	31.8	0	30.8	0
No diabetes	6,271 / 3,643	33.0	+5	32.2	+4	30.6	0	30.3	+1
Diabetes	953 / 590	50.3	−30	49.4	−30	39.0	−1	33.7	−4
Below 70 mg/dL	102 / 58	59.2	+59	58.0	+57	51.3	+49	62.4	+44
70 to 180 mg/dL	6,661 / 3,800	25.2	+6	24.6	+5	23.3	+4	24.5	+4
Above 180 mg/dL	461 / 375	100.2	−93	98.2	−91	86.4	−74	63.6	−44

[a] Readings without / with calibration (the calibrated model scores only readings after a patient's first). Diabetes: pre-operative diabetes in the VitalDB case table (363 patients). Below 70 mg/dL: 102 readings from 75 patients without calibration.

Table S5. MUST, matched comparison on the 48 recordings (22 people) usable by every model; all models trained on VitalDB. $\Delta\text{RMSE} = \text{RMSE}_{\text{model}} - \text{RMSE}_{\text{constant}}$; a positive value means the model is less accurate than the constant prediction.

Model	RMSE (mg/dL) [95% CI]	MARD (%)	Pearson r	ΔRMSE vs constant [95% CI]
PPG features (GBM)	23.3 [17.2, 29.4]	14.8	−0.01	+2.6 [−1.0, +6.9]
PPG features + age, sex, BMI (GBM)	22.4 [16.2, 28.3]	13.7	0.17	+1.6 [−2.4, +4.9]
Age, sex, BMI (GBM)	21.8 [15.5, 28.6]	12.7	0.33	+1.0 [−3.6, +5.0]
ResNet34-1D, person split	21.2 [16.4, 26.1]	14.9	−0.22	+0.4 [−0.2, +1.0]
TinyCNN, person split	21.6 [17.5, 25.8]	15.8	−0.03	+0.9 [−0.2, +2.1]
Constant: VitalDB training mean	20.7 [16.2, 25.6]	14.7	–	–

Table S6. Sensitivity of the optical results to the tissue model: best glucose-equivalent error (mg/dL).

Configuration	Main model (Table 12)	Continuous model	Continuous, glucose also in fat: water-weighted	Continuous, glucose also in fat: full strength
Single wavelength, best channel	71	73	89	68
Oximeter wavelengths, 660 / 940 nm	82 / 84	87 / 91	102 / 129	136 / 114
Glucose absorption only, best channel	790	790	750	625
Pulsatile channel below 1000 nm	no signal[a]	no usable signal[b]	no signal[a]	no signal[a]
Two wavelengths	none	none	none	none
Three wavelengths	none	none	>1000	>400
Four / five / six wavelengths[c]	11 / 11 / 11	15 / 13 / 13	20 / 19 / 18	18 / 17 / 16
All wavelengths (idealized bound)	4.0	5.1	5.6	4.8

none: glucose cannot be separated from the confounders. Glucose also in fat: the dermal glucose sensitivities are applied to the subcutaneous fat as well, either scaled by the water fraction of fat relative to dermis (0.22/0.65 = 0.34, water-weighted) or unscaled (full strength); both variants add the confounders of the fat layer (its water, blood volume and scattering).

[a] Glucose absorption is set to zero below 1000 nm in this configuration, so the pulsatile channel carries no glucose signal there by construction.

[b] Zero by construction without glucose absorption below 1000 nm; with the weak absorption of Haque et al. [13] the glucose-equivalent error is about 4.9×10^5 mg/dL (the signal is not strictly zero, but far too weak to use). The other rows are identical with and without this absorption to the precision shown.

[c] Best sets of wavelengths at least 30 nm apart; Monte Carlo 95% intervals within ±1 mg/dL.

Table S7. ESP32-S3 resources and timing for one 60-s window (240 MHz). The self-test times one stored window on the board at start-up; the hardware-in-the-loop value is the median over 30 different recordings streamed from the PC. Processing time depends on the number of beats in a window, so the two differ slightly.

Item	Value	Agreement with Python
Filtering and beat extraction	274.7 ms	identical beat counts
15 device features	32.7 ms	bit-identical
300 trees, calibration-free	3.1 ms	0 mg/dL
300 trees, calibrated	3.5 ms	0 mg/dL
Device model, total (self-test)	310.6 ms	0 mg/dL
Device model, hardware-in-the-loop, median (range), 30 recordings	304 (253–340) ms	0 mg/dL
Per-beat TinyCNN, 84 beats (network 1,271.2 ms)	1,554.7 ms	7.6×10^{-6} mg/dL
Flash: firmware + two tree models (8,700 nodes each)	945 kB	
Free heap after start-up (largest block)	80 kB (30 kB)	

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